Introduction to Mechanical and Electrical Engineering and Automation

AUTO2001E

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STUDY GUIDE

The prescribed text books/papers for this course are:

(1) ***Vector Mechanics for Engineers: Dynamics****, Tenth Edition, by F.P. Beer, E.R. Johnston, and P.J. Cornwel., McGraw-Hill Publications.*

(2) ***Physics for Scientists and Engineers****, Sith Edition, by Tripler, P.A., Mosca, G., Mcmillan Education,*

(3) *Alam, M.M., "**A Novel Parent-Offspring Framework for Identifying Vectors and Scalars in Mechanics**," International Research Journal of Advanced Engineering and Science, Volume 11, Issue 2, pp. 101-116, 2026.*

STUDY GUIDE

LECTURE SCHEDULE

Week	Day	Period	Methods	Content covered
4	Thursday 24 Sept	3-4	Lecture	Fundamental of Mechanical Engineering
5	Tuesday 29 Sept	3-4	Lecture	Parameters in Mechanical Engineering
6	Thursday 8 Oct	3-4	Lecture	Forces, work and energy
7	Tuesday 13 Oct	3-4	Lecture	Static and dynamic Equilibrium

STUDY GUIDE

Class Test

One class test: 13 October (about 25 mins)

Syllabus: class lectures on 24 Sept, 29 Sept, and 8 Oct

Mechanical Engineering problems



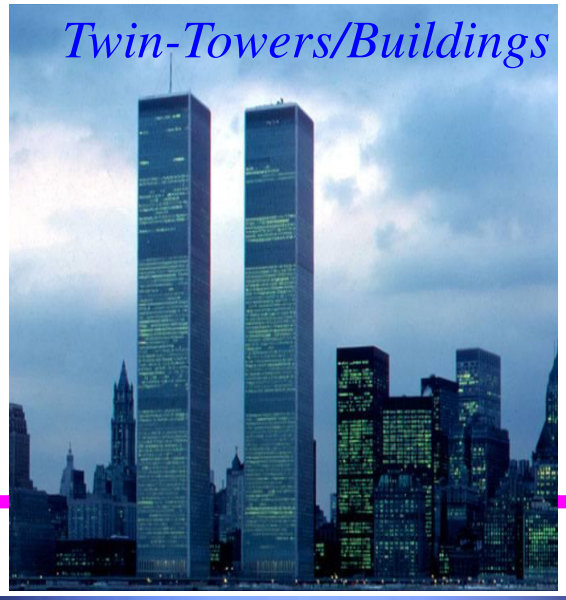
Gunagzhou Humen bridge: May 5, 2020

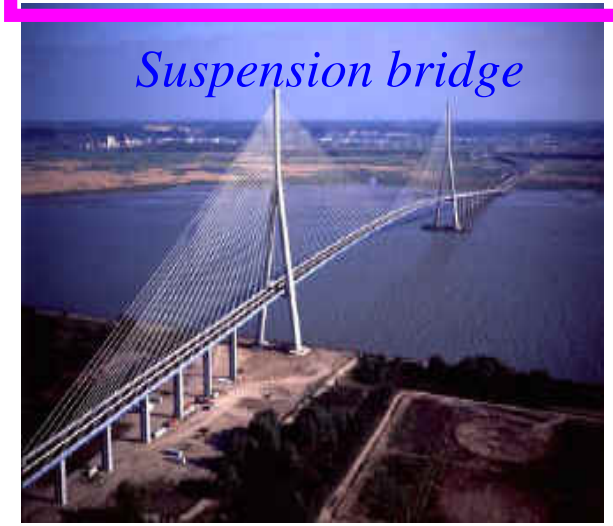
Mechanical Engineering problems



Tacoma Bridge Failure

Mechanical Engineering problems







Mechanical Engineering problems





Mechanical Engineering problems



Wind turbine tower vibration

Mechanical Engineering problems



Mechanical Engineering problems



Photo 11.2 Multiple cables and pulleys are used by this shipyard crane.

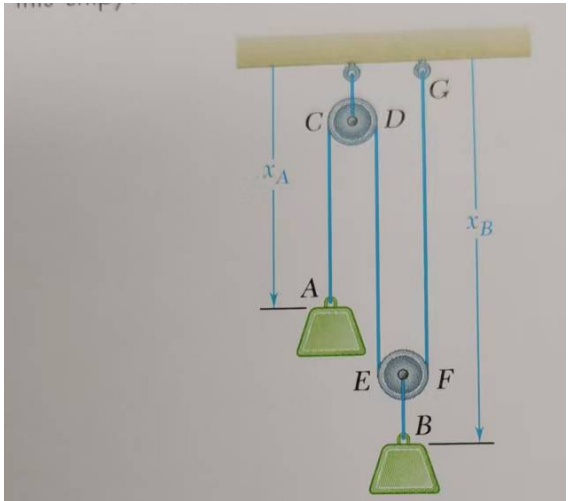


Fig. 11.8

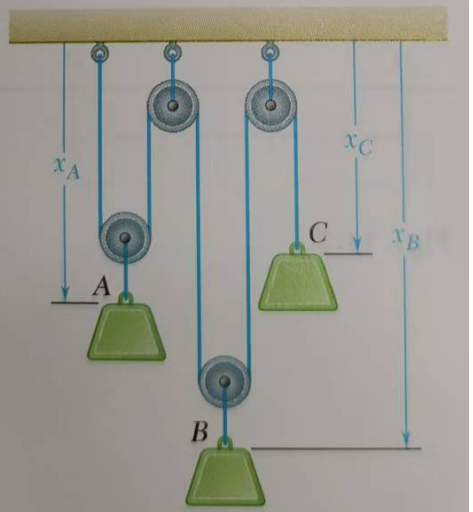


Fig. 11.9

Mechanical Engineering problems

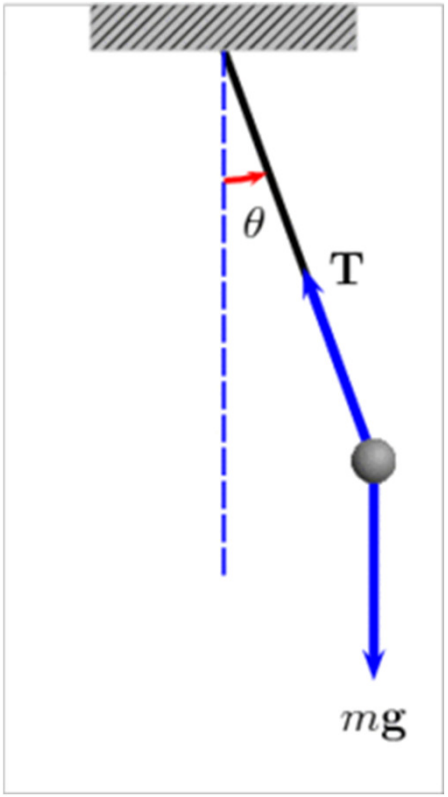


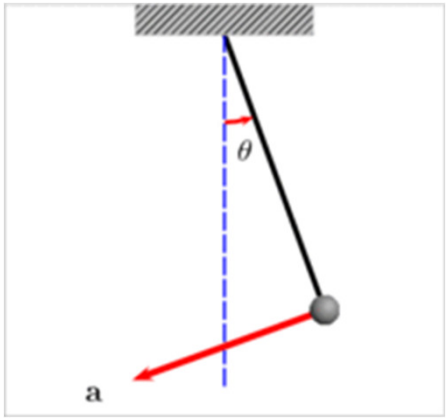




Mechanical Engineering problems



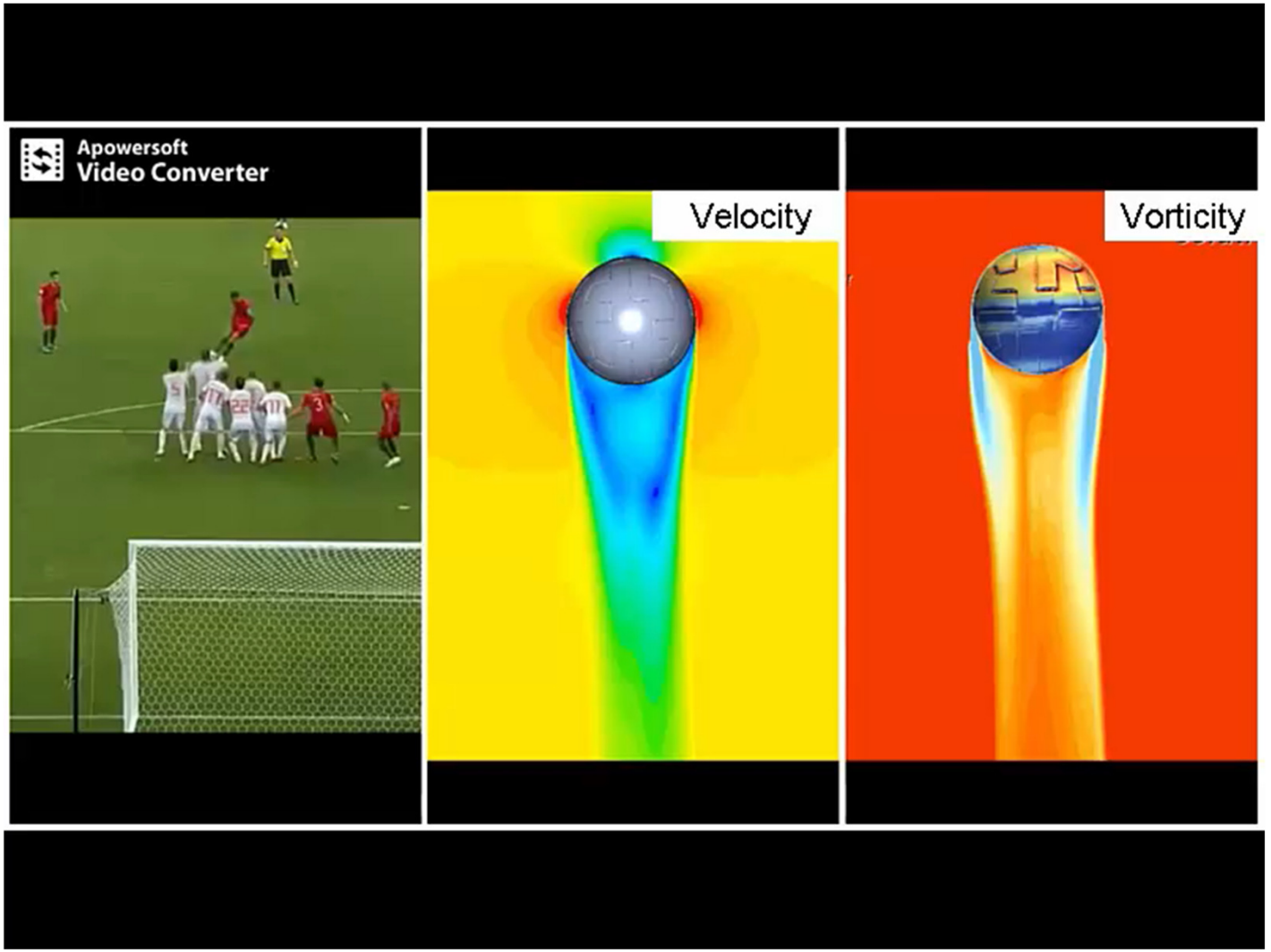




Mechanical Engineering problems



Mechanical Engineering problems



Mechanical Engineering problems

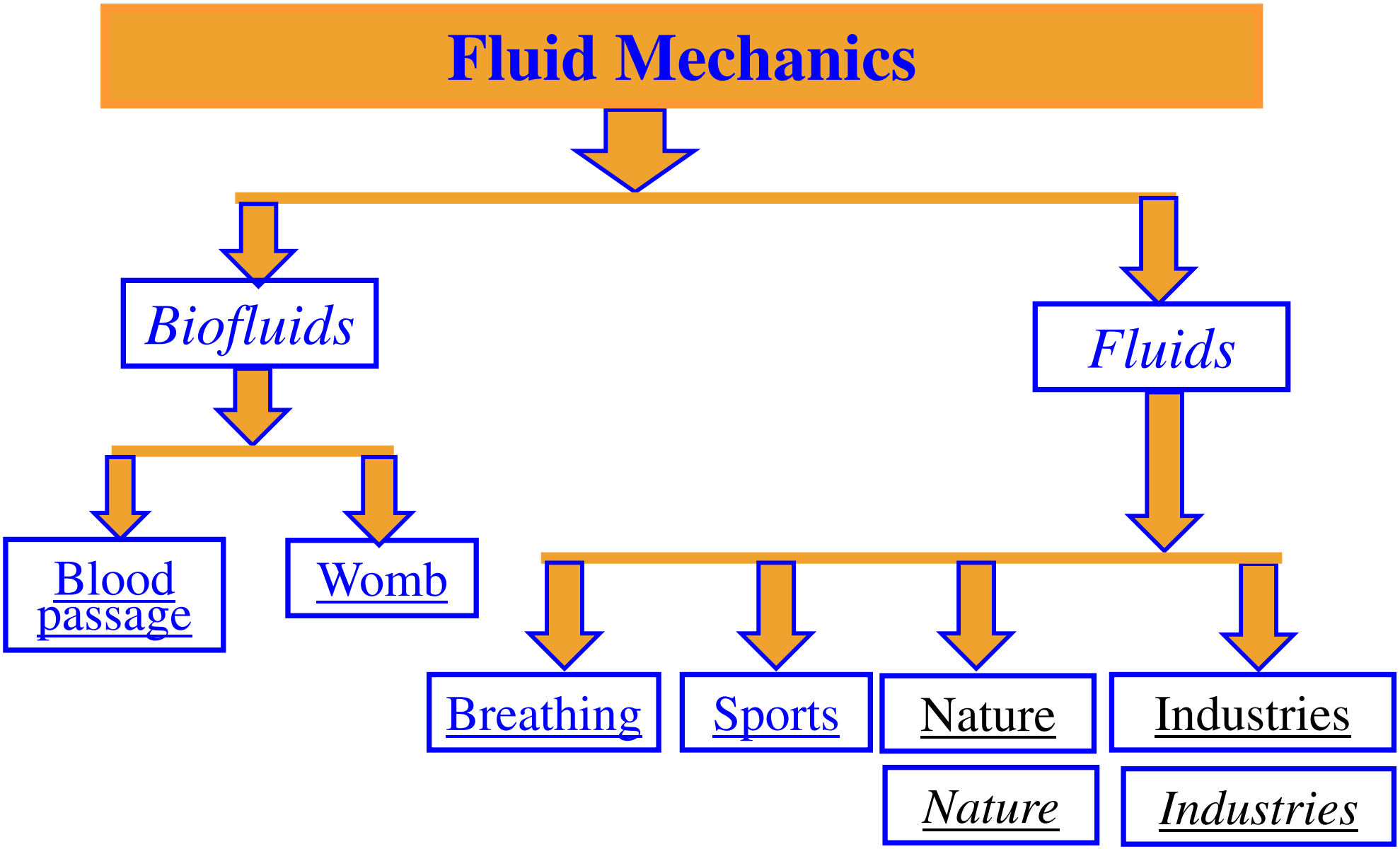
Free Vortex: Tornado



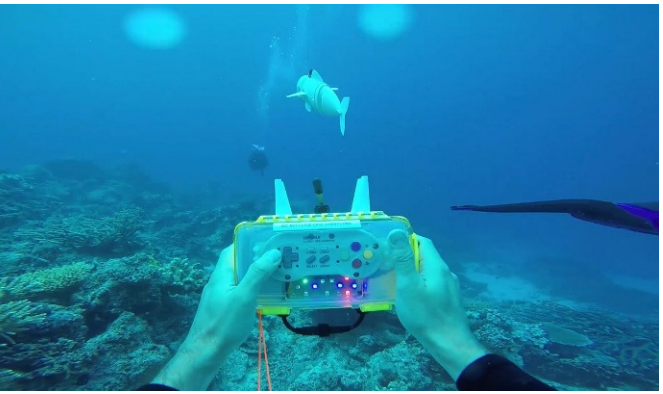




Mechanical Engineering problems



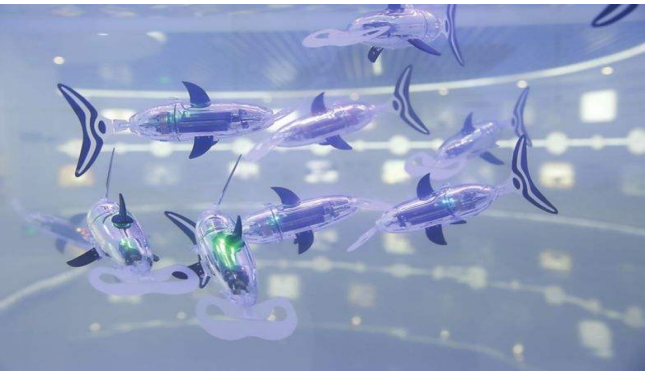
Mechanical Engineering problems



Coastal Survey



Coastal defense and Environmental Protection



Robot fish



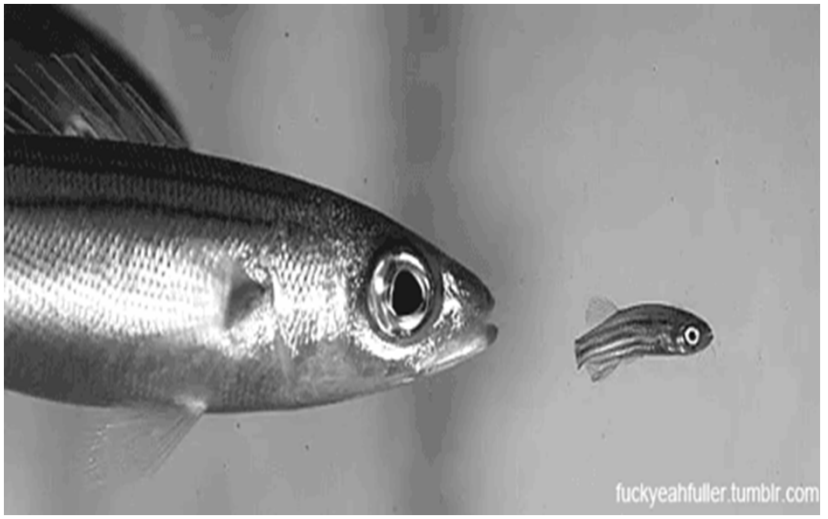
Bio-Mimic Robot

Applications of underwater robots

- *Deep-sea mining*
- *Coastal security*
- *Monitoring ecosystems*
- *Spy missions*
- *Surveillance missions*
- *Inspecting offshore drilling rigs*
- *Monitoring fish farms*
- *Tracking pollution*

Mechanical Engineering problems





Thrust is the most important factor for prey-predator competition

The hydrodynamic origin of thrust generation must be understood to explain

(i) How a quick start (agility) is possible to escape from a predator

(ii) How to get backward movement

Mechanical Engineering problems







What fluid dynamics is to be known for such a jumping robot?

Mechanical Engineering problems



Travelling Wave (TW)

Anguiliform

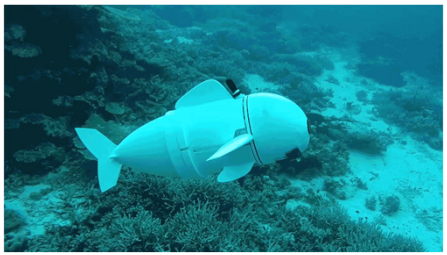




Pitching+TW

Carangiform

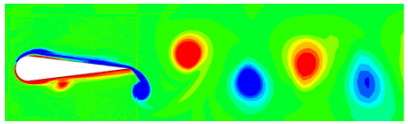






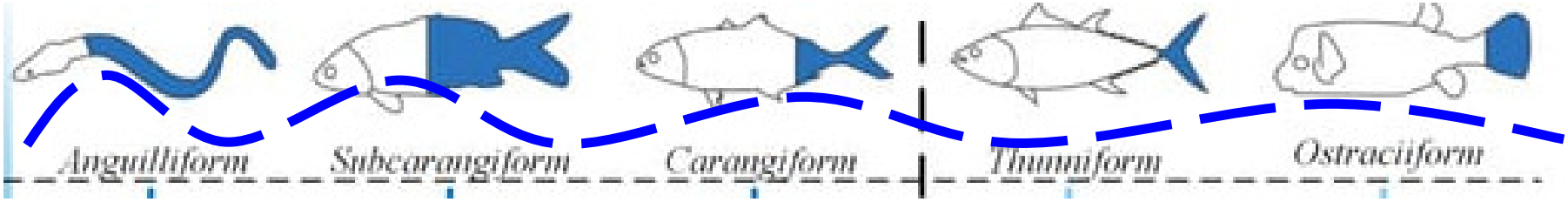
Pitching

Thunniform





Native swimmers → *Hydrodynamic model* → *Engineering applications*



Hydrodynamics of real swimmers may be useful for designing bio-inspired swimming robot.

Mechanical Engineering problems

Can fish swim backward? Did you see it?

Ans: Yes, most fish can swim backward, but not all (e.g. Shark, Tuna)



Courtesy: YouTube

Black ghost fish often swim backward

Shark



Tuna



They can have high speed

Why are they unable to move backward?

Mechanical Engineering

Mechanics is the heart of *mechanical, civil, electrical, aerospace, or even biomedical engineering*

Fundamental principles are the "heart of mechanics": *Engineering mechanics, Fluid Mechanics, Solid Mechanics.*

Parameters?

Parameters

A **parameter** is a quantity or variable that defines certain characteristics or behavior of a system, process, or model.

Importance: Parameters are essential for understanding and predicting the behavior of physical systems. They help define the conditions under which a system operates and how it responds to different influences

Classical Mechanics: Fundamental parameters and derived parameters

Mathematics: Fundamental/prime numbers and composite numbers

Language: Vowels and consonants

Natural Numbers

***Definition**: The set of positive integers used for counting.*

***Examples**:* 1,2,3,4,5, ...

***Note**: Sometimes,* 0 *is included in the set of natural numbers, but traditionally, natural numbers start at* 1.

Natural Numbers

Prime numbers

A prime number is defined as a natural number (greater than 1) that has only two factors: **1 and itself**.

***Examples**:* 2, 3, 5, ...

Composite numbers

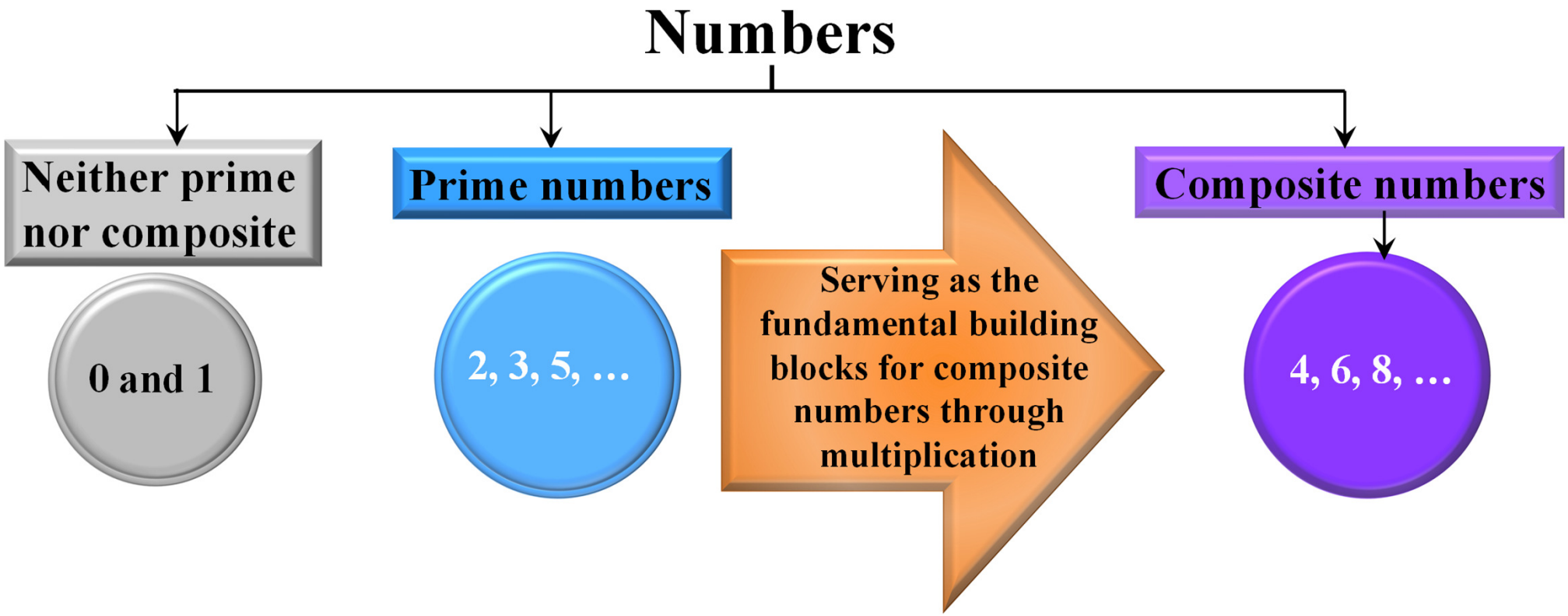
A **composite number** is a natural number (greater than 1) that has **more than two factors**

***Examples**:* 4, 6, 8 ...

*Composite numbers are made up of **prime factors**.*

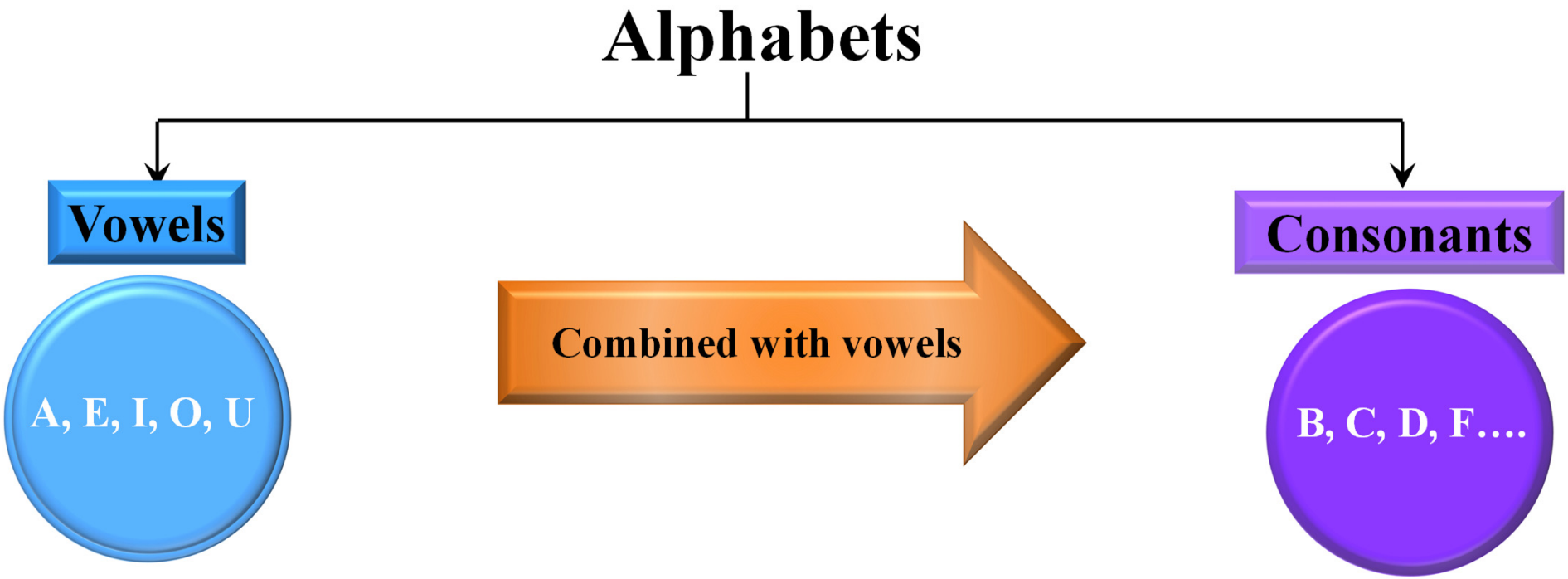
Prime numbers are 'Building Blocks of composite numbers

Natural Numbers



Properties of prime and composite numbers

Alphabets



Properties of vowels and consonants

Classical Mechanics

Parameters

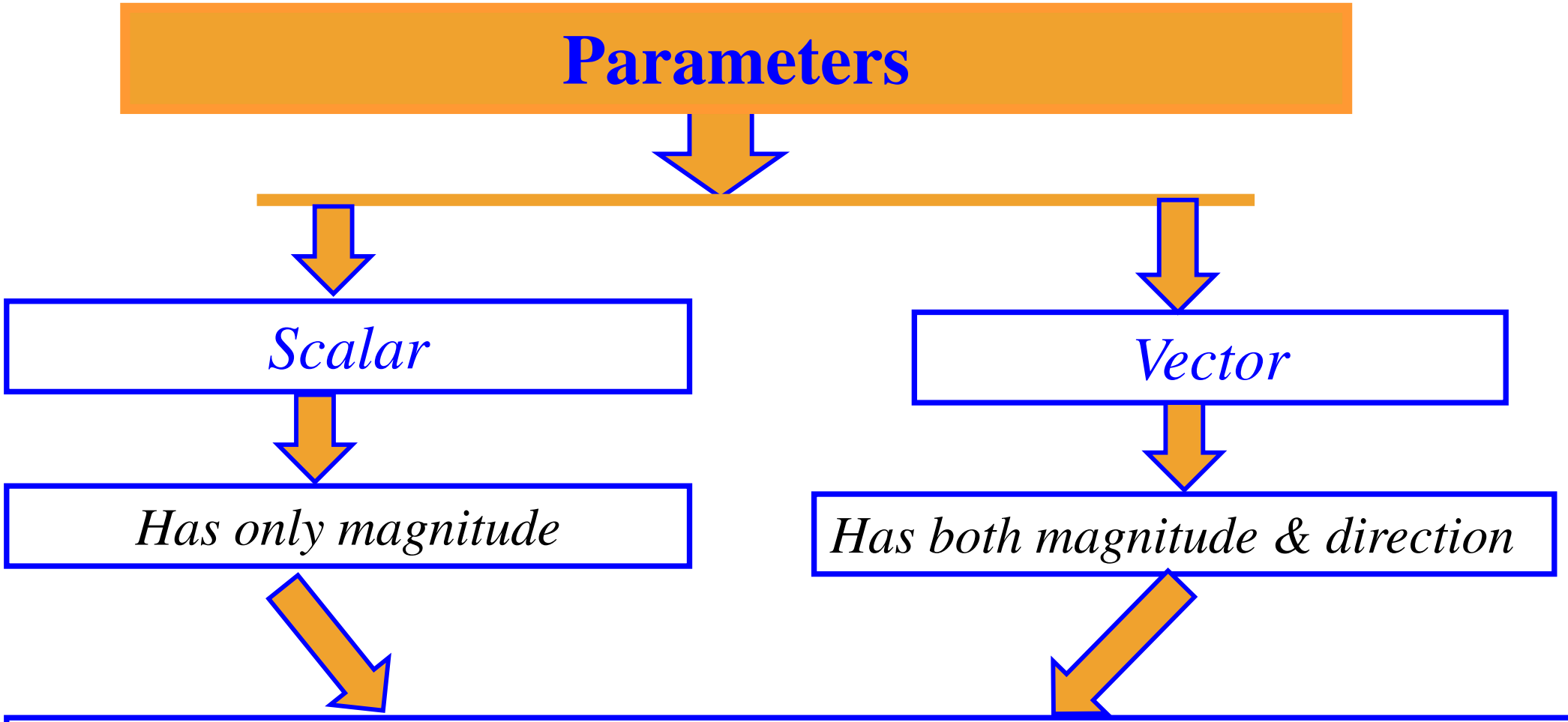
Fundamental parameters

Length, mass, and time are fundamental because they are the basic units and cannot be defined by other quantities

Derived parameters

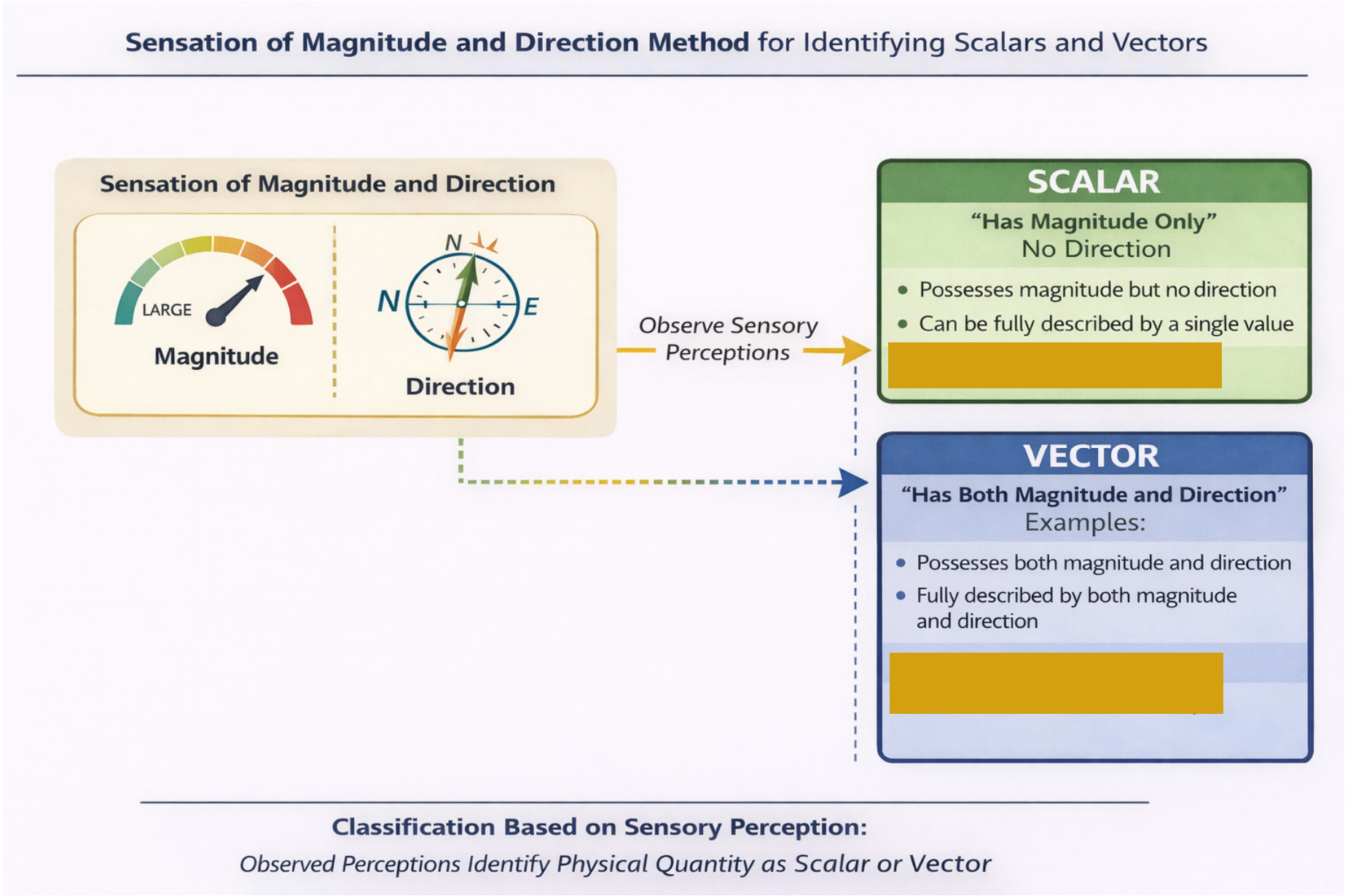
- ***Derived from** fundamental parameters*
- ***Combinations** of fundamental quantities.*
- *Describe more complex physical phenomena.*

Classical Mechanics



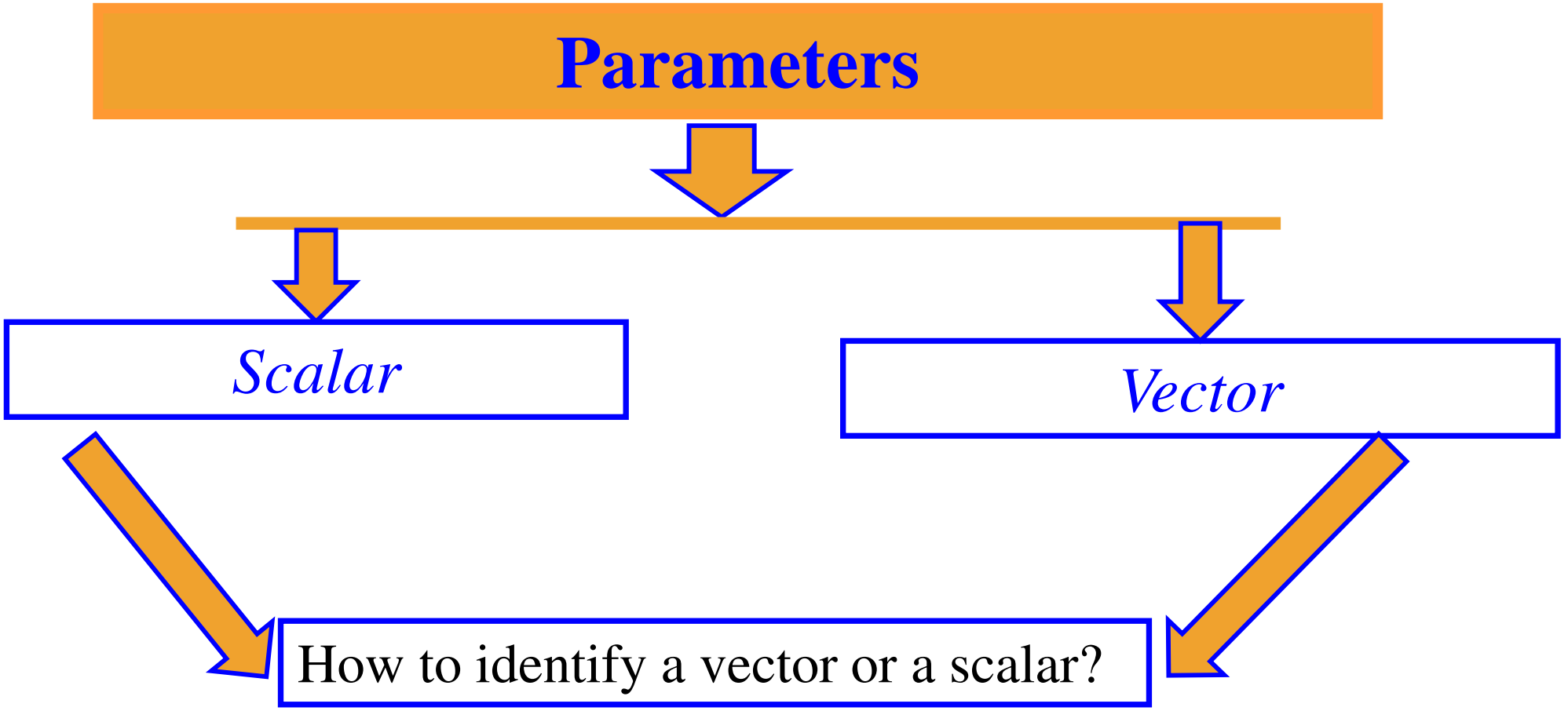
Identifying whether a quantity or parameter is a ***vector*** *or a* ***scalar*** *is crucial in physics, engineering, and mathematics because it dictates* ***how we handle the quantity*** *in calculations,* ***how it behaves in different situations****, and* ***how we apply laws of physics*** *to solve problems.*

Classical Mechanics



Identification of scalars and vectors based on the sensation of magnitude and direction

Classical Mechanics



Classical Mechanics

Humans

Male

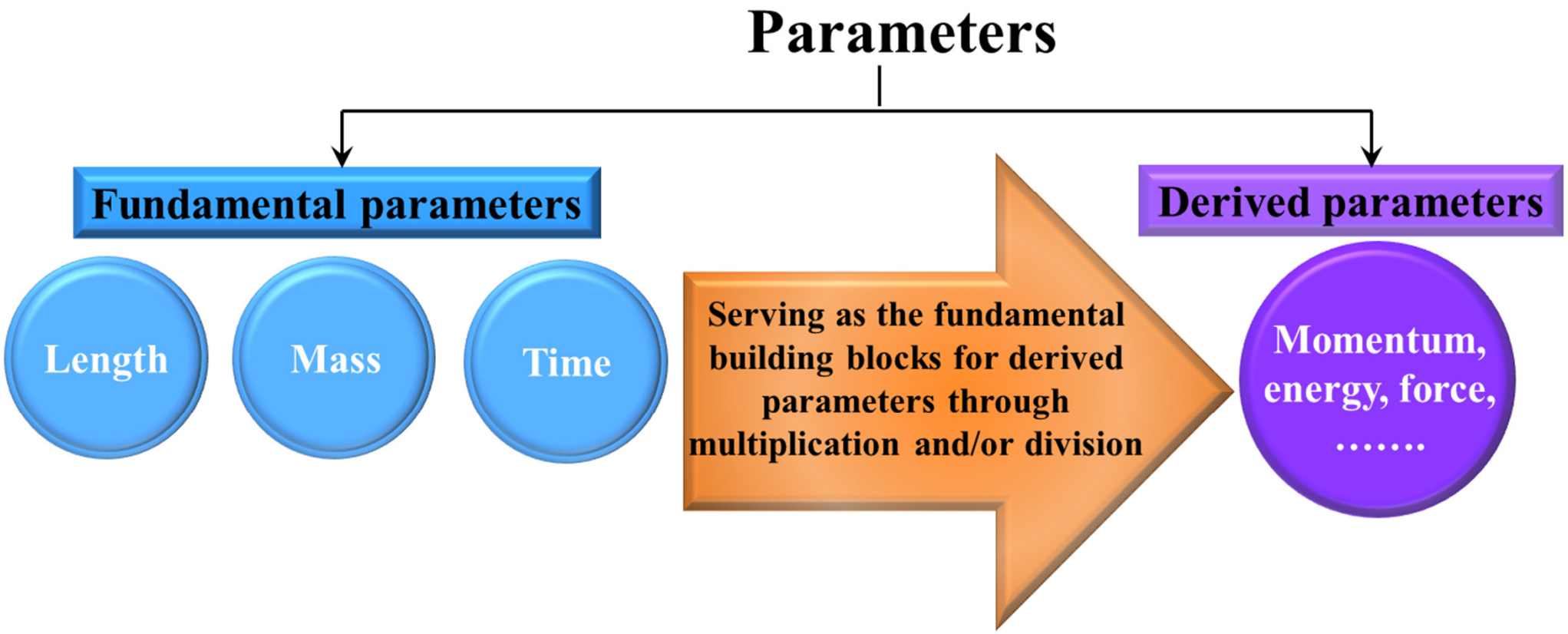
Female

How to identify a male or a female?

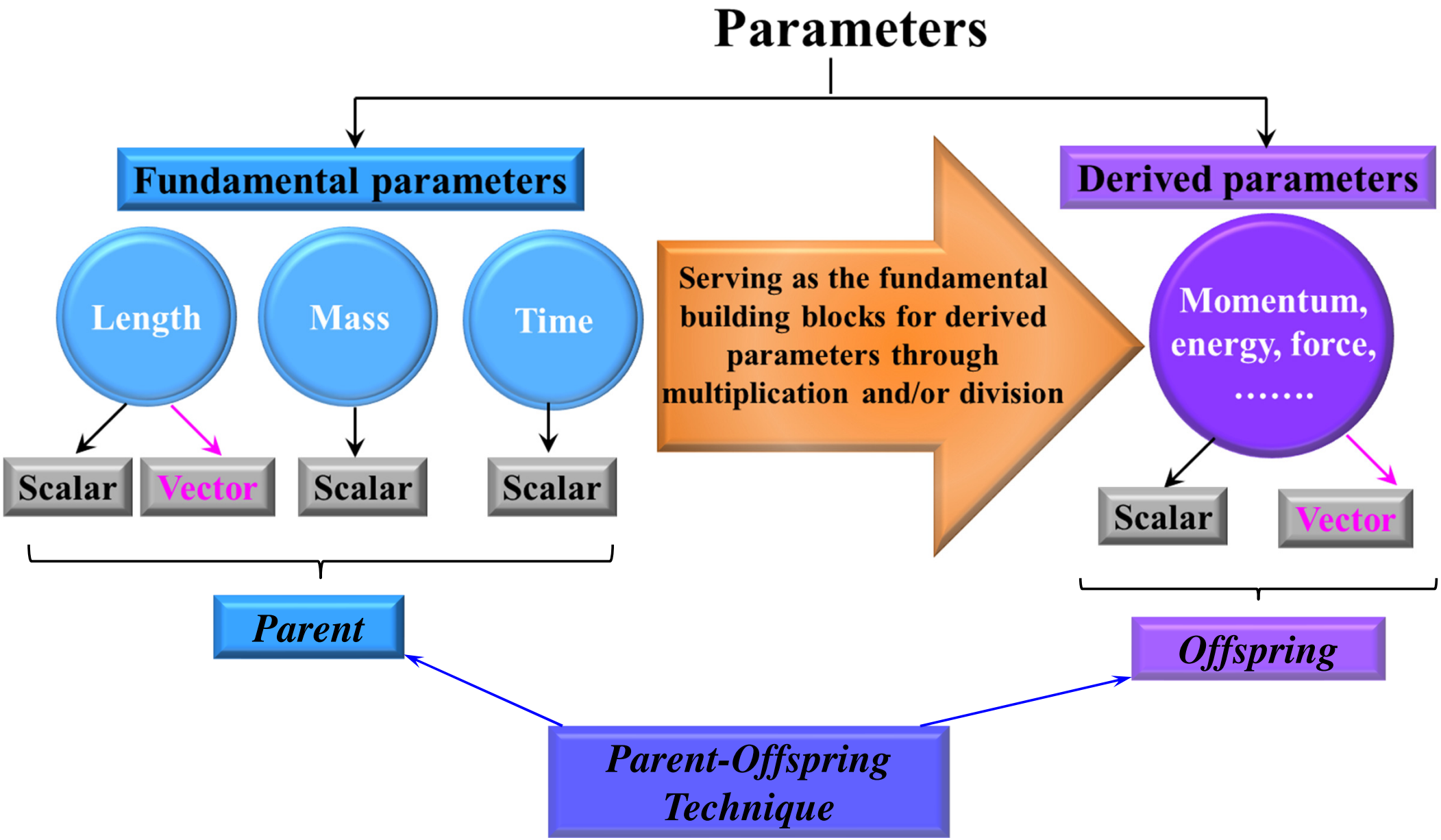


Identification based solely on outward perception is unreliable

Classical Mechanics



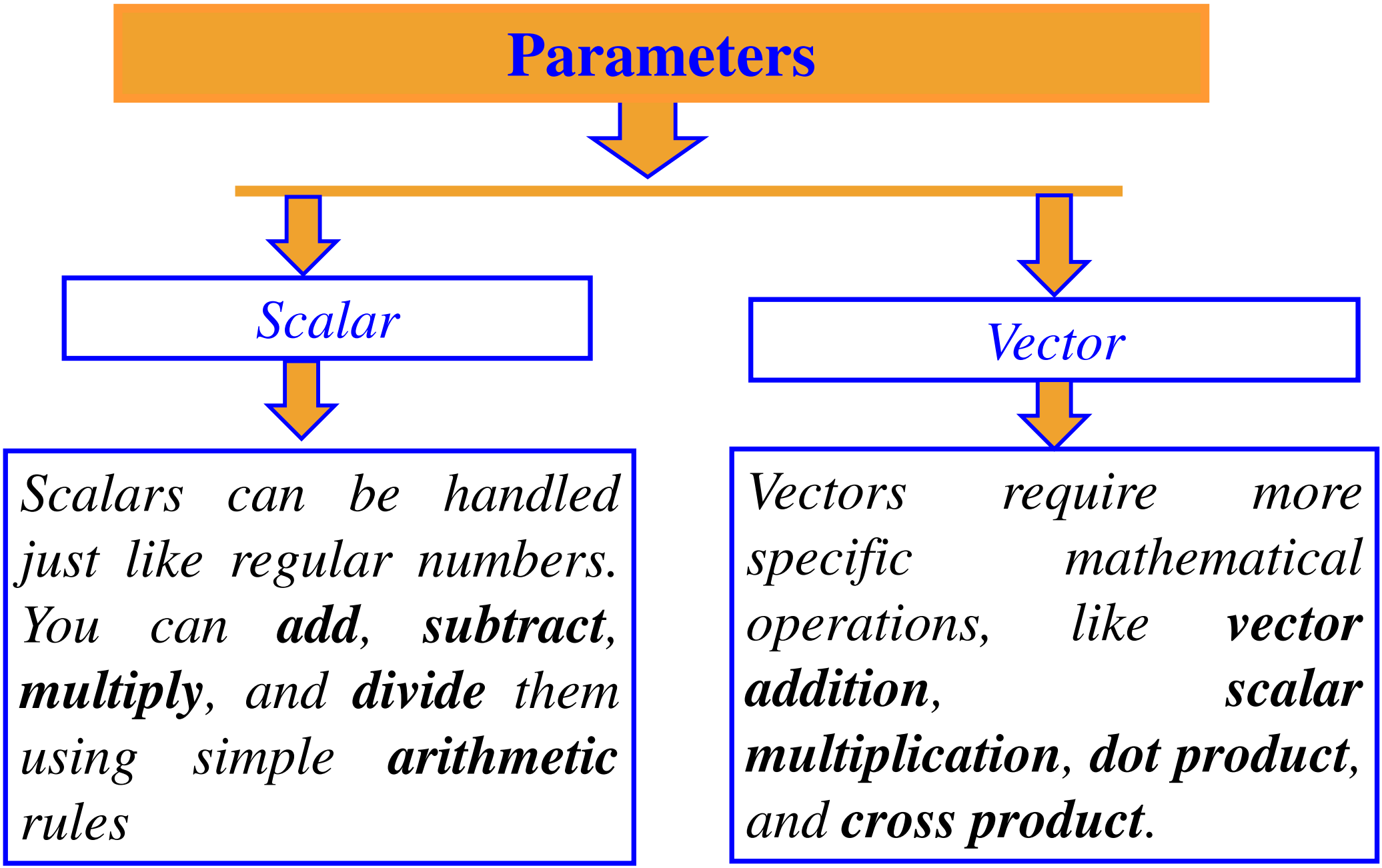
Classical Mechanics



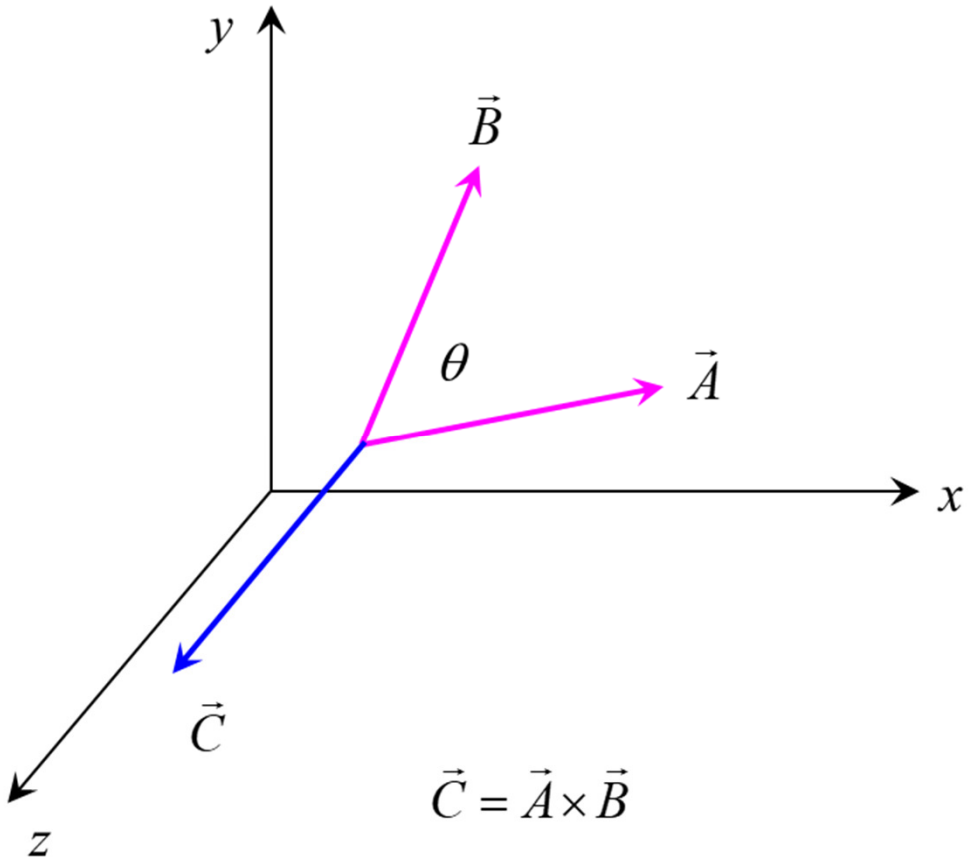
Classical Mechanics

Approaches → Properties ↓	Magnitude & direction	Parent-offspring
Description	Observing the physical properties of a quantity	Classifying quantities based on relationship between derived ('offspring') quantities and fundamental ('parent') quantities
Scalars/Vectors Identification	Scalars: magnitude only Vectors: magnitude plus direction	Scalars/Vectors are identified by how offspring inherit properties from parents
Advantages	Intuitive and simple for basic quantities	Provides hierarchical, physically motivated classification; useful for derived quantities
Limitations	Fails for derived or complex quantities; may not capture transformation behavior	Requires knowledge of the derivation or dependency relationships; may be more complex to implement

Classical Mechanics

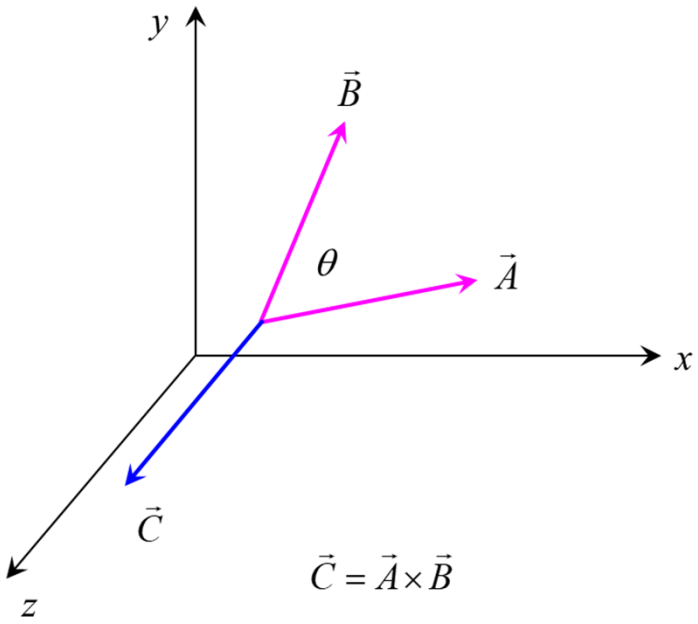


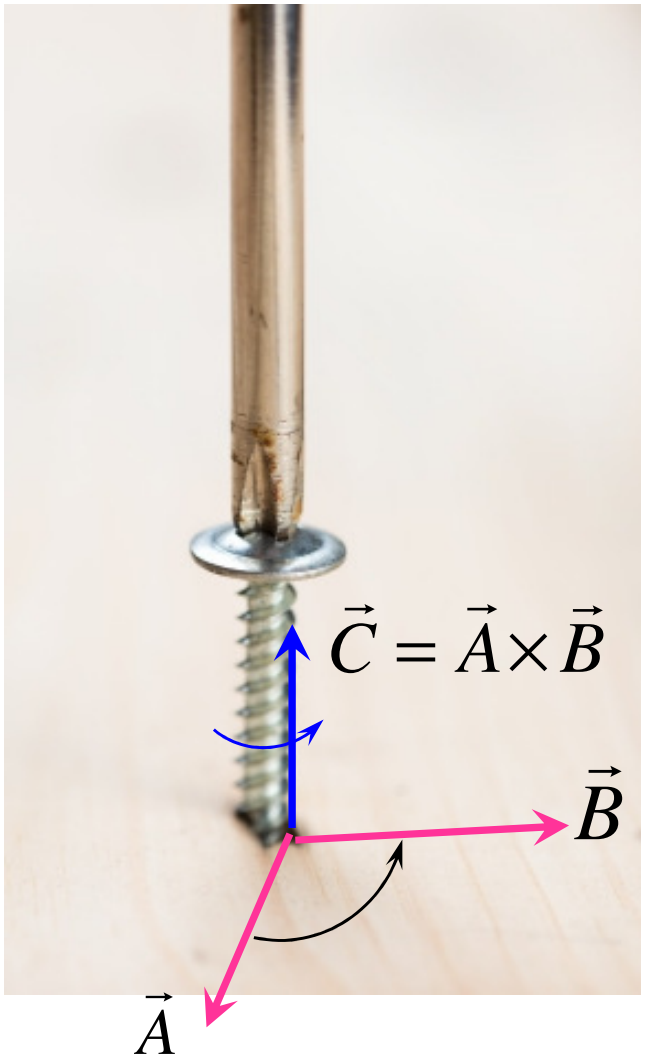
Classical Mechanics

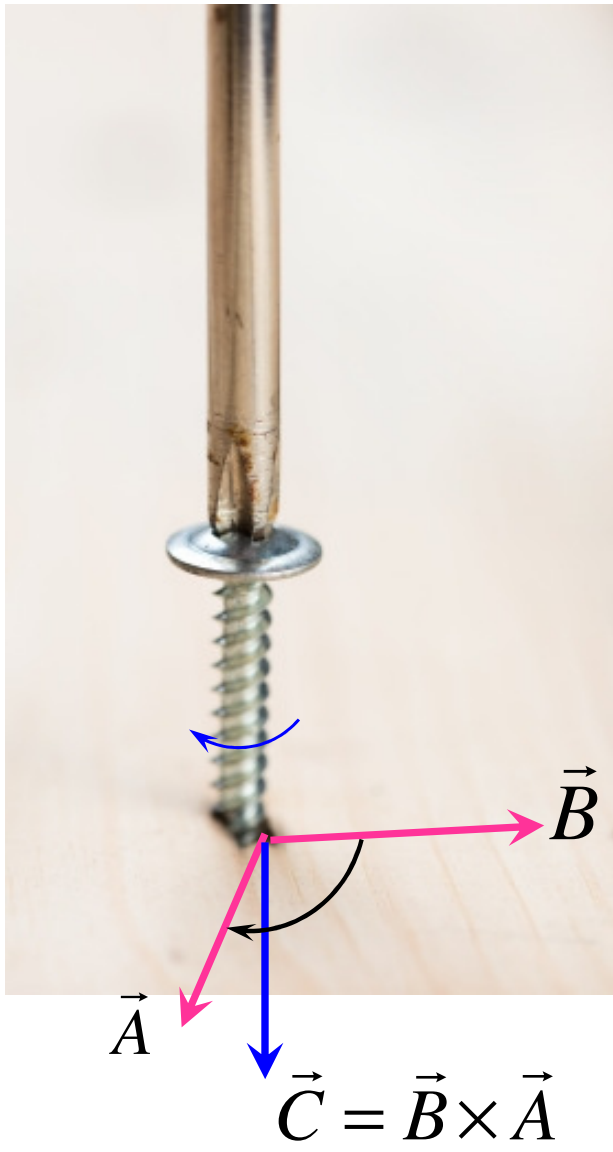


Vector operations: cross-product

Classical Mechanics



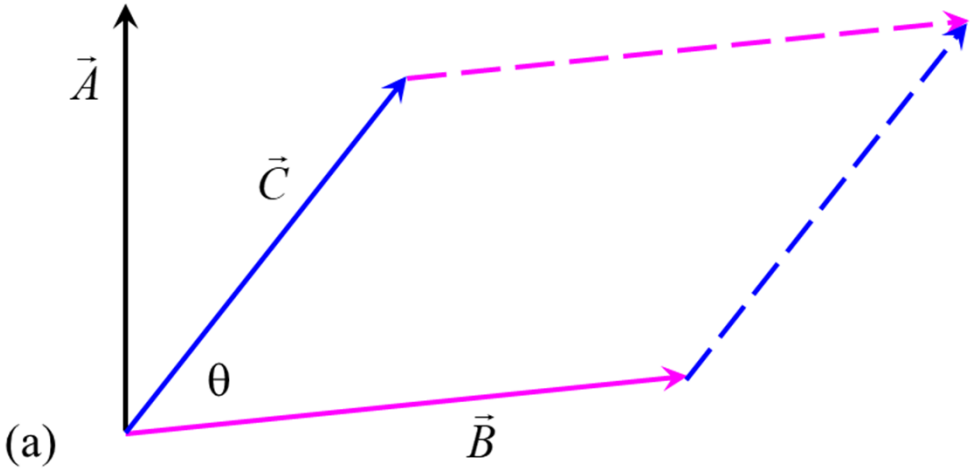




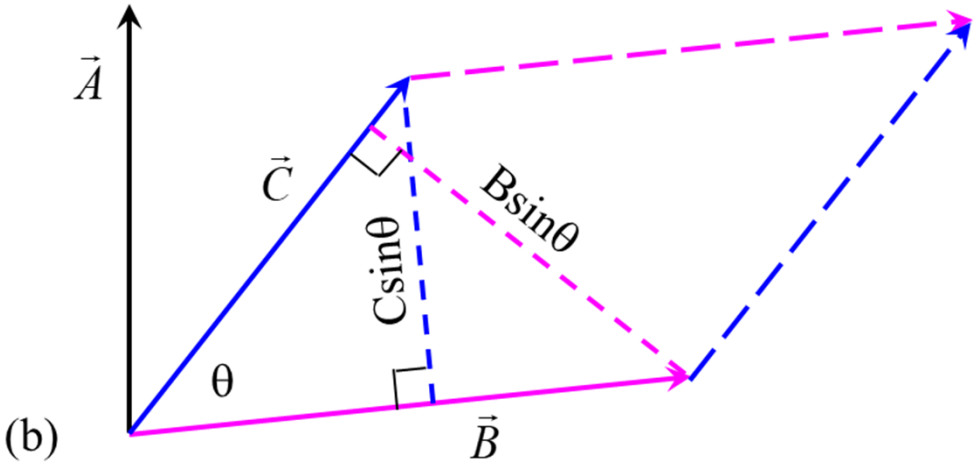
Cross-product: **Determining Direction from Screw Rotation — An Easy Technique**

Classical Mechanics

Is area a vector or a scalar?



$$\vec{A} = \vec{B} \times \vec{C}$$

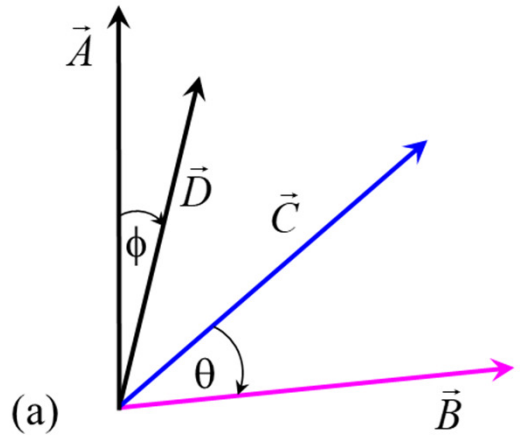


$$A = BC\sin\theta$$
$$= B|C\sin\theta|$$
$$= C|B\sin\theta|$$

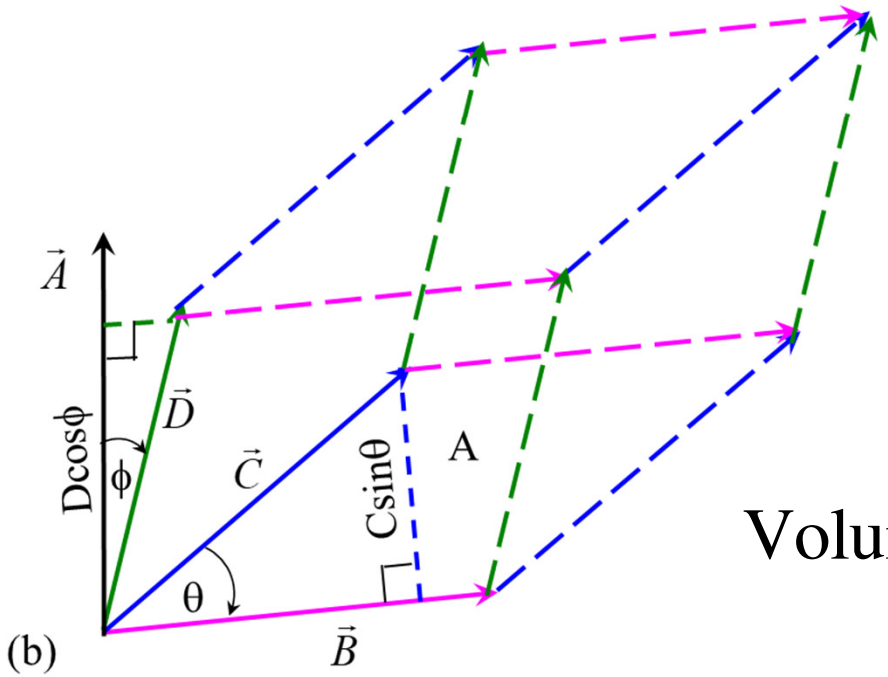
Representation of area as a vector obtained from the cross product of two vectors.

Classical Mechanics

Is volume a vector or a scalar?



$$\vec{A} = \vec{B} \times \vec{C}$$



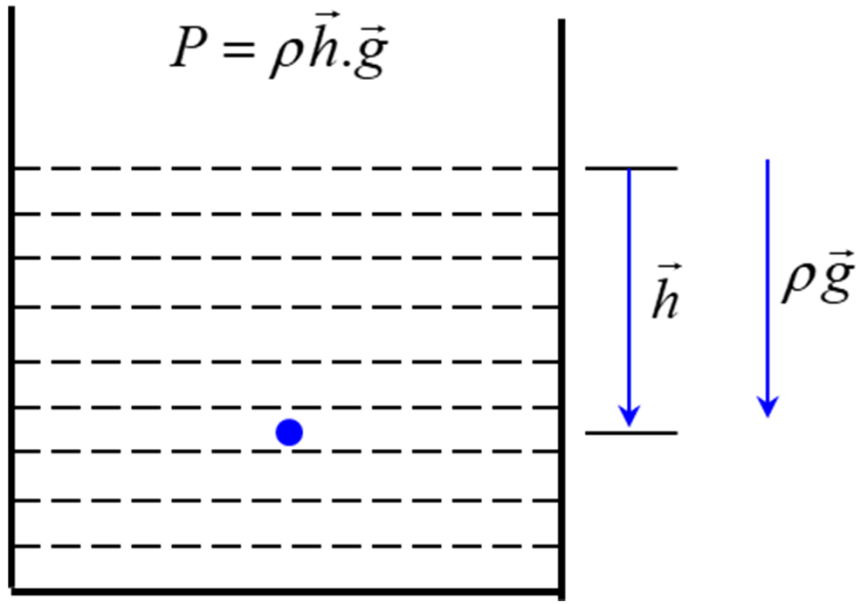
$$\text{Volume} = \text{ADcos}\phi = \vec{A}.\vec{D} = (\vec{B} \times \vec{C}).\vec{D}$$

$$\text{Volume} = (\vec{B} \times \vec{C}).\vec{D} = (\vec{D} \times \vec{B}).\vec{C} = (\vec{C} \times \vec{D}).\vec{B}$$

Representation of volume as a scalar obtained from the cross and dot product of three vectors.

Classical Mechanics

Is pressure a vector or a scalar?



Representation of pressure P as a scalar quantity, obtained from the dot product of height and gravitational acceleration

Reference: *Alam, M.M., "A Novel Parent-Offspring Framework for Identifying Vectors and Scalars in Mechanics," International Research Journal of Advanced Engineering and Science, Volume 11, Issue 2, pp. 101-116, 2026.*

Classical Mechanics

Parameters

Dimensional

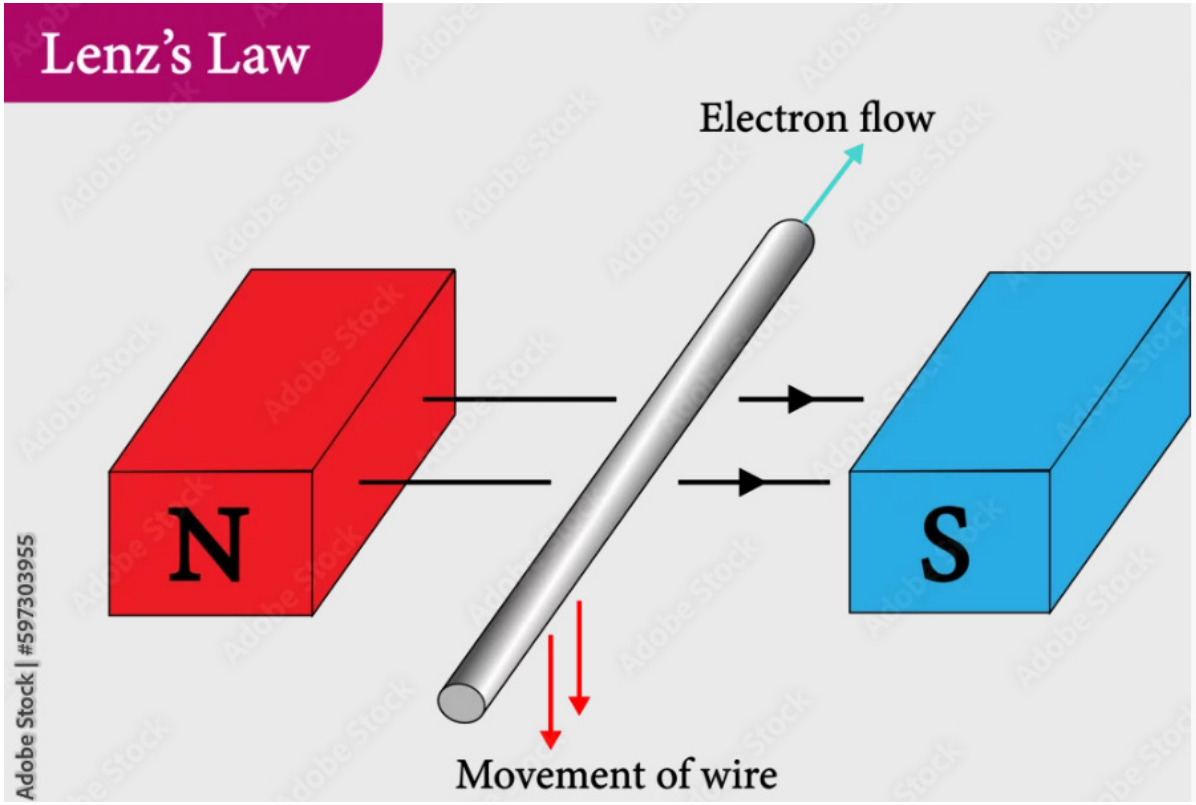
Dimensional parameters have ***units***

Non-dimensional

- *Non-dimensional parameters have no* ***units***
- *They result from the* ***ratio of two quantities*** *with the same dimensions or units*

Classical Mechanics

Force, Moments, and their relationship with EMF

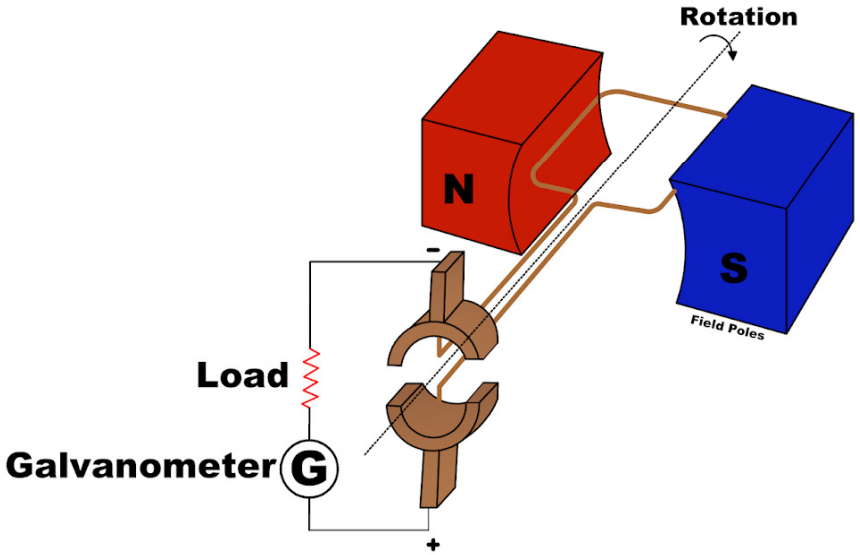


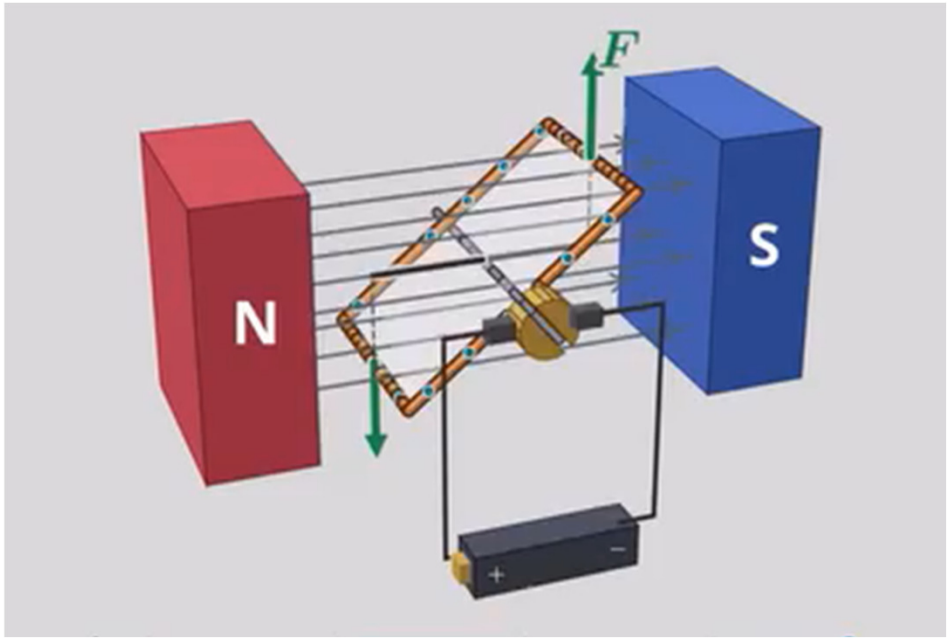
Lenz's law specifies the **direction** of the induced current

Classical Mechanics

Force, Moments, and their relationship with EMF

Faraday's Law of Electromagnetic Induction





Force and energy

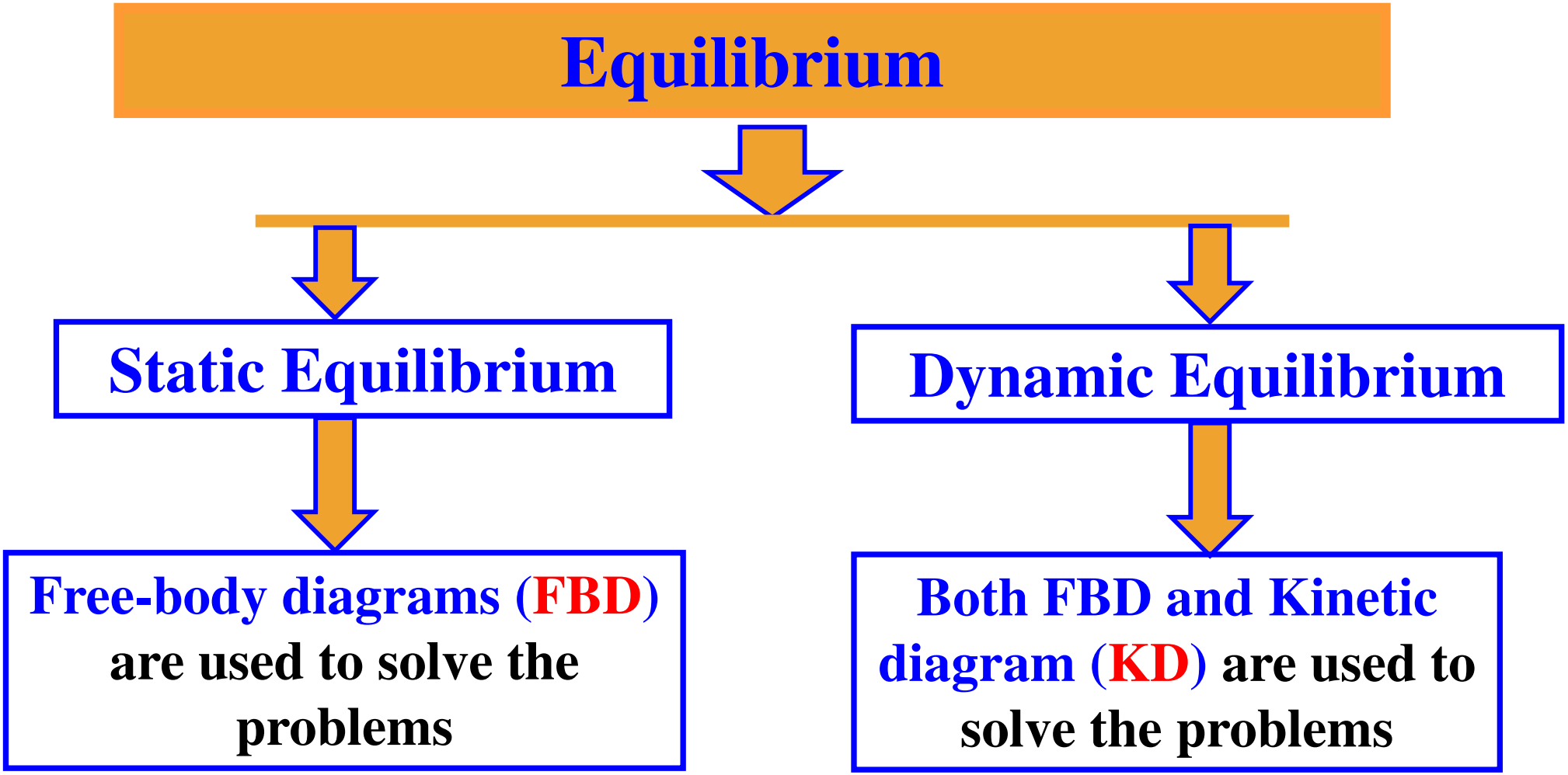
Position vector
and
relative position vector





Equilibrium

Equilibrium



Equilibrium

Dynamics or motion

Kinetics

Relationship between motion and force



Kinematic

Relationship between displacement, velocity, acceleration and time without reference to cause of motion

